openKVM v1 — KVM-over-IP ESP32-P4

Schema preliminaire rev 0.1 — revue humaine obligatoire avant fabrication.

Chaque net est porte par une etiquette globale : memes noms = meme connexion.

Alimentation

USB-C, OU ideal 5 V, buck 3,3 V, buck 1,2 V

File: power.kicad_sch

ESP32-P4

SoC, flash, quartz, DCDC coeur, boutons, LED, debug

File: mcu.kicad_sch

Capture HDMI

Connecteur HDMI, ESD, TC358743, oscillateur 27 MHz

File: hdmi.kicad_sch

Ethernet

PHY IP101GRI, quartz 25 MHz, RJ45 a magnetiques integres

File: eth.kicad_sch

Peripheriques

microSD (UHS-I), ecran SPI, ESD USB

File: periph.kicad_sch

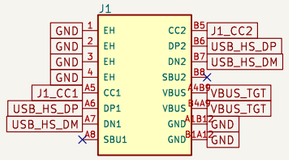
Genere par hardware/scripts/gen_kicad.py depuis design.py
Schema preliminaire : revue humaine obligatoire avant fabrication
openKVM — CERN-OHL-S-2.0

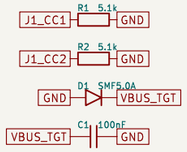
Sheet: /
File: openkvm.kicad_sch

Title: openKVM v1 — vue d ensemble

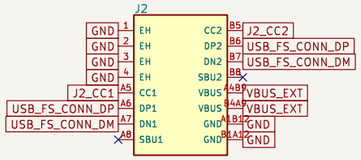
Size: A4	Date: 2026-11-12	**Rev: 0.1**
KiCad E.D.A. 9.0.9		Id: 1/6

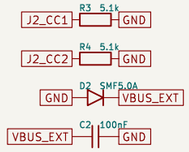
USB-C CIBLE



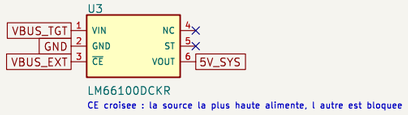


USB-C EXT

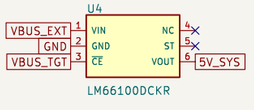


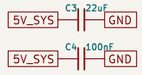


OU ideal 5 V



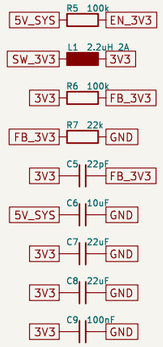
CE croisee : la source la plus haute alimente, l autre est bloquee





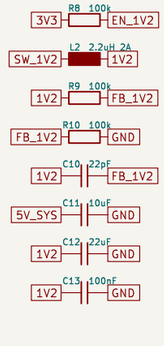
Buck 3,3 V





Buck 1,2 V (TC358743)





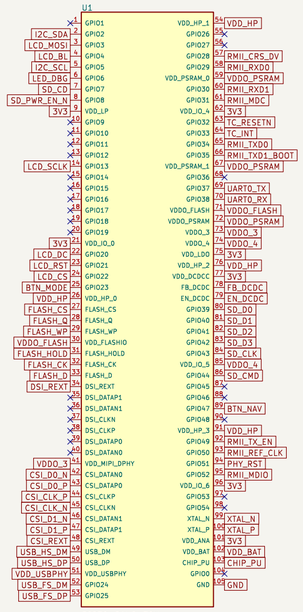
Genere par hardware/scripts/gen_kicad.py depuis design.py
Schema preliminaire : revue humaine obligatoire avant fabrication
openKVM — CERN-OHL-S-2.0

Sheet: /Alimentation/
File: power.kicad_sch

Title: openKVM — Alimentation

Size: A3	Date: 2026-11-12	**Rev: 0.1**
KiCad E.D.A. 9.0.9		Id: 2/6

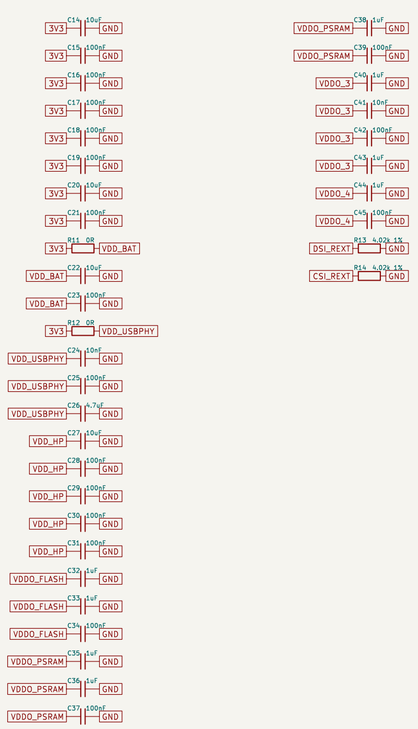
ESP32-P4



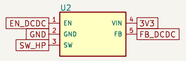
ESP32-P4NRW32X

revision de puce >= 3.x obligatoire

Decouplage P4 (au plus pres des broches)

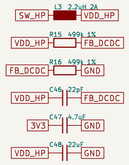


DCDC coeur (pilote par le P4)



TLV62569DBVR

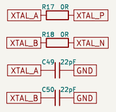
modele valide par Espressif : pas de variante a decharge de sortie



Quartz 40 MHz



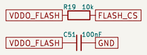
40MHz 10ppm



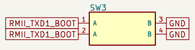
Flash SPI 16 Mo



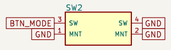
GD25Q128BESIGR

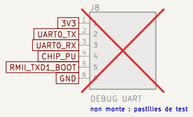


Reset, boot, boutons, LED

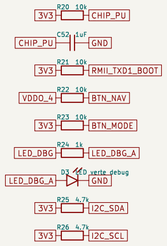








non monte : pastilles de test

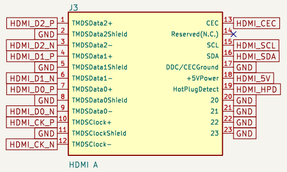


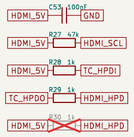
Genere par hardware/scripts/gen_kicad.py depuis design.py
Schema preliminaire : revue humaine obligatoire avant fabrication
openKVM - CERN-OHL-S-2.0
Sheet: /ESP32-P4/
File: mcu.kicad_sch

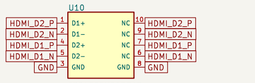
Title: openKVM - ESP32-P4

Size: A2	Date: 2026-11-12	**Rev: 0.1**
KiCad E.D.A. 9.0.9		Id: 3/6

Connecteur HDMI



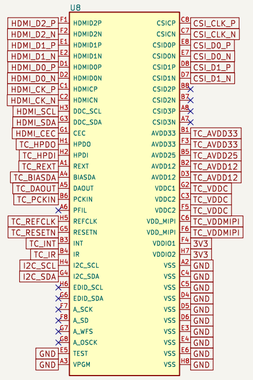


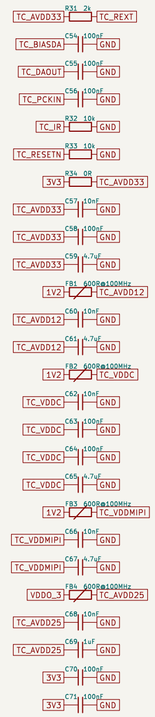




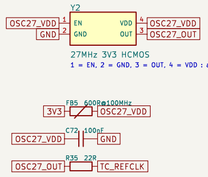


TC358743XBG



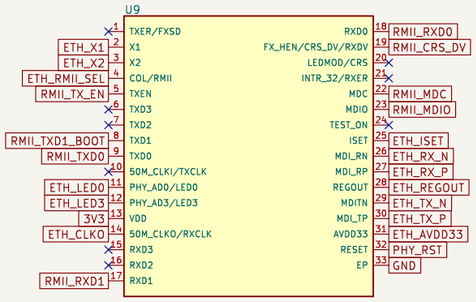


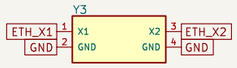
Oscillateur 27 MHz

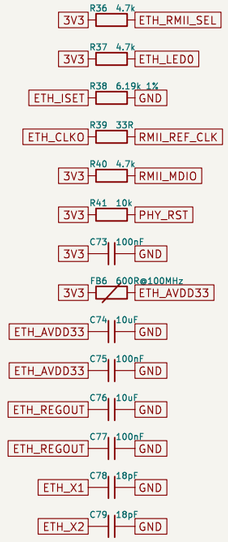


Genere par hardware/scripts/gen_kicad.py depuis design.py
Schema preliminaire : revue humaine obligatoire avant fabrication
openKVM — CERN-OHL-S-2.0
Sheet: /Capture HDMI/
File: hdmi.kicad_sch
Title: openKVM — Capture HDMI
Size: A2 | Date: 2026-11-12 | **Rev: 0.1**
KiCad E.D.A. 9.0.9 | Id: 4/6

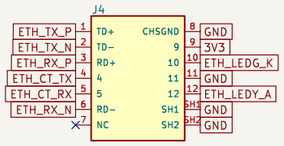
PHY IP101GRI

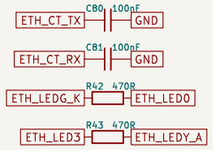






RJ45





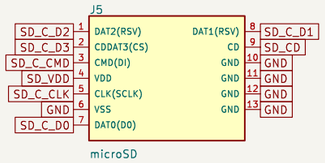
Genere par hardware/scripts/gen_kicad.py depuis design.py
Schema preliminaire : revue humaine obligatoire avant fabrication
openKVM — CERN-OHL-S-2.0

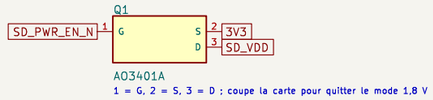
Sheet: /Ethernet/
File: eth.kicad_sch

Title: openKVM — Ethernet

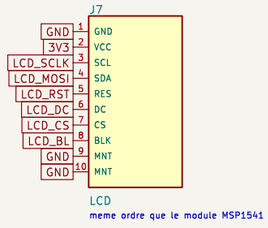
Size: A3	Date: 2026-11-12	**Rev: 0.1**
KiCad E.D.A. 9.0.9		Id: 5/6

microSD (slot SDMMC 0, UHS-I)

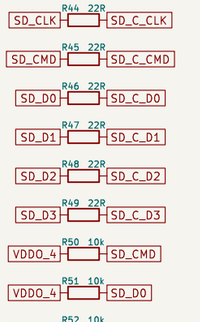




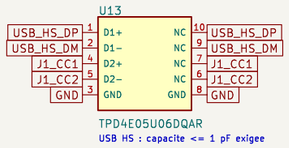
Ecran ST7789 240x240

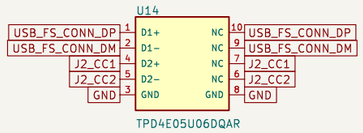


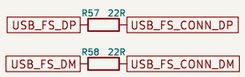




ESD USB







Genere par hardware/scripts/gen_kicad.py depuis design.py
Schema preliminaire : revue humaine obligatoire avant fabrication
openKVM — CERN-OHL-S-2.0

Sheet: /Peripheriques/
File: periph.kicad_sch

Title: openKVM — Peripheriques

Size: A3	Date: 2026-11-12	**Rev: 0.1**
KiCad E.D.A. 9.0.9		Id: 6/6